

CS & IT ENGINEERING



Computer Network

Introduction

Lecture No. - 02



By - Abhishek Sir



Recap of Previous Lecture



Topic

Concepts of Layering

Topic

OSI & TCP/IP Model





Topics to be Covered



Topic

Application Layer

Topic

Two Process Communication

Topic

Transport Layer

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : TCP/IP Model



Application Layer

Transport Layer

Internet Layer

Link Layer



Topic : OSI Model



Layer 7

Application Layer

Layer 6

Presentation Layer

Layer 5

Session Layer

Layer 4

Transport Layer

Layer 3

Network Layer

Layer 2

Data Link Layer

Layer 1

Physical Layer

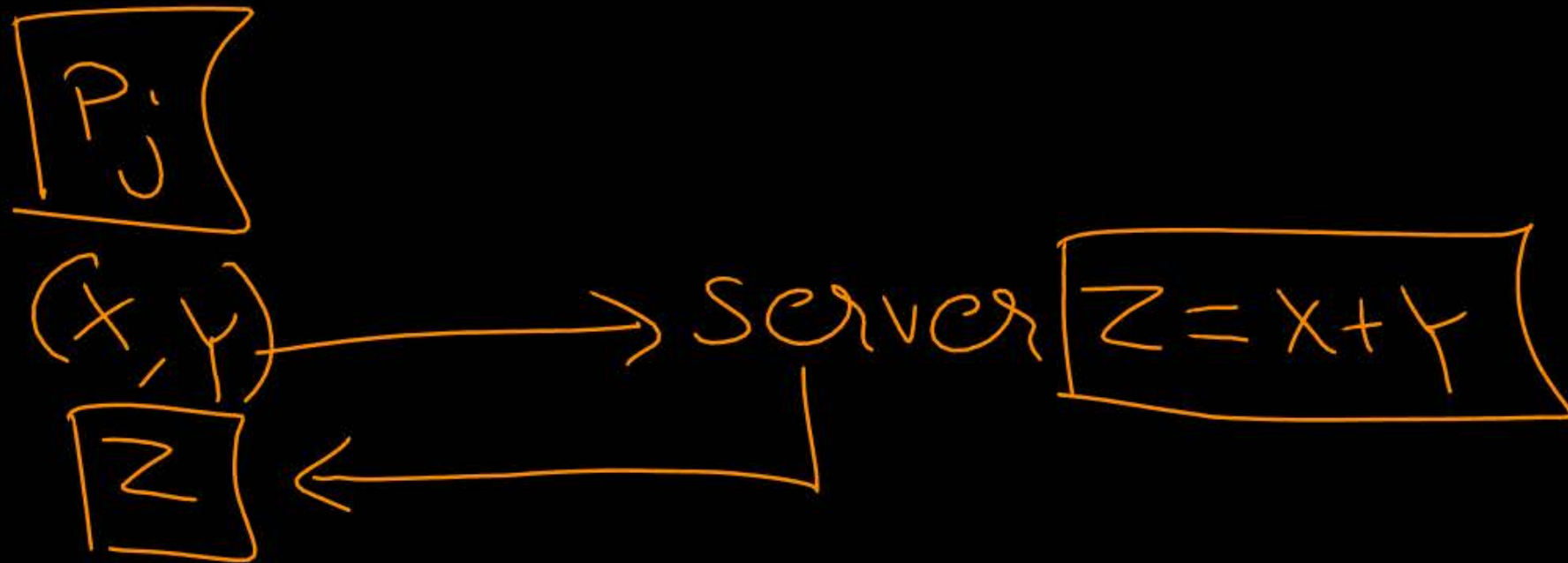
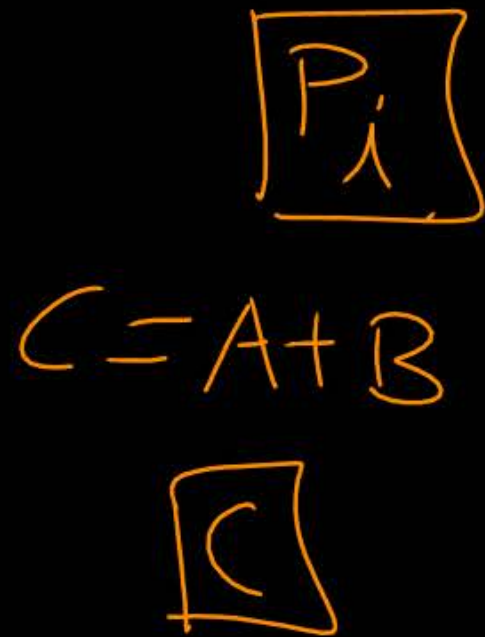


Topic : Application Layer



→ Network application program (network process)
[e.g. Browser]

→ **Process** : Program in execution
[Process id (pid) managed by OS]





Topic : Port Number



- Port Number (16-bit) : Identifier 2 byte
- Used to identify a network process in a machine
[Process involved in Network Communication]

16-bit
Unsigned
Range

$0 \text{ to } (2^{16} - 1)$

Total = 2^{16} numbers

- Port Number managed by OS
[different with process id (pid)]

* Port no. should be unique
within a machine

⇒ Two (network) process running in a machine
can't have same port no. at a time.



Topic : Port Number



→ Port numbers are divided into three ranges :

1. System (Well-known) Ports : 0 to 1023
[assigned and controlled]
 2. Registered Ports : 1024 to 49151
[neither assigned nor controlled,
but can be registered to prevent duplication]
 3. Dynamic (Private) Ports : 49152 to 65535
[neither assigned nor controlled nor registered]
- Server Process
- Client Process



Topic : Two Process Communication

Two network processes belongs to :

1. Same Host

→ Communicate using IPC (Inter-process communication) of OS

2. Different Hosts

→ Communicate over network 



Topic : Protocol



- Set of Rules
- Define, how data is exchanged between network processes





Topic : Application Layer



Application Layer Protocols :

- DNS : Domain Name System
- HTTP : Hyper Text Transfer Protocol
- FTP : File Transfer Protocol
- SMTP : Simple Mail Transfer Protocol



Topic : PDU



=> Protocol Data Unit (PDU)

→ Basic unit of exchange ←

→ Between same protocols of different machine

→ Between 'Layer-n protocol' of one machine
and 'Layer-n same protocol' of other machine



Topic : Application Layer



Application Layer PDU : "Message"

#Q. The protocol data unit (PDU) for the application layer in the Internet stack is:

[GATE-2012, 1-Mark]

IIT-D

(A) Segment

(B) Datagram

✓ (C) Message

(D) Frame

Ans: C



Topic : Two Process Communication



- Two network processes, belongs to different hosts
- Communicate over network, by exchanging "messages"



Topic : Two Process Communication



Two paradigm :

1. Client - Server

[e.g. Web browsing, Email]

2. Peer to peer [client-client]

[e.g. BitTorrent, VoIP]



Topic : Client - Server Paradigm



Server :

- Always on host → [24/7 running]
- [Permanent IP Address]
- Mostly in data centers

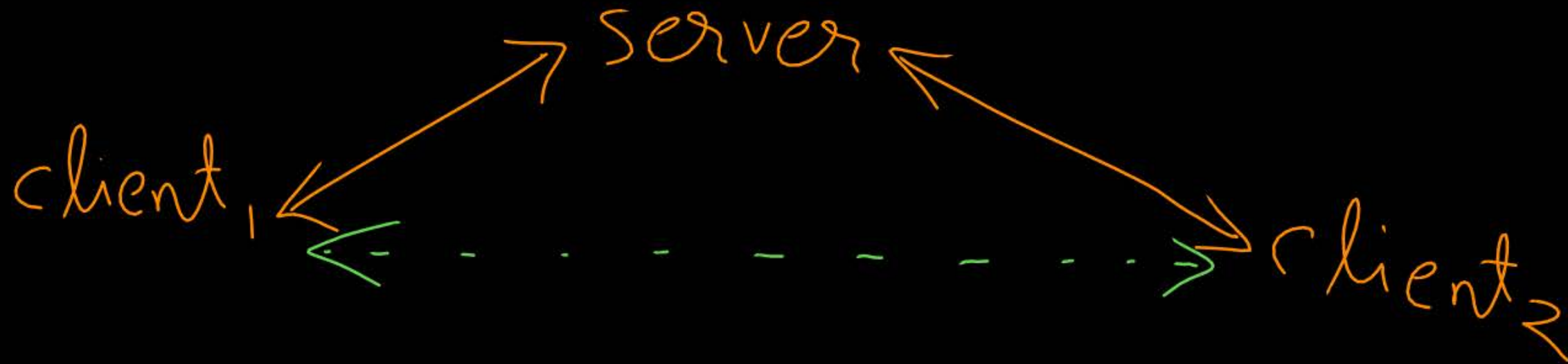


Topic : Client - Server Paradigm



Client :

- Communicate (contact) with server
- May have dynamic IP address
- Clients do not communicate directly with each other





Topic : Client - Server Paradigm



Example :-

1. Web browsing

Client : Web browser

Server : Web Server

2. Email

Client : User Agent

Server : Mail Server



Topic : Peer - Peer Architecture



- No always on server
- Arbitrary end systems directly communicate
- Complex management (Peers can change IP address)
- e.g P2P file sharing (BitTorrent), Voice over IP (VoIP) →

→ Two client can communicate directly

#Q. Which one of the following is not a client server application?

[GATE-2010, 1-Mark]

IIT-G

- ~~(A) Internet chat~~ → client-server
- ~~(B) Web browsing~~ → web client - Web Server
- ~~(C) E-mail~~ → Mail Client - Mail Server
- ✓ (D) Ping

Ans: D

#Q. Which one of the following is not a client server application?

[GATE-2010]

- (A) Internet chat
- (B) Web browsing
- (C) E-mail
- (D) Ping

Ans : (D) Ping

Ping is utility, to check connectivity between either client - client or client - server.



Topic : Two Process Communication

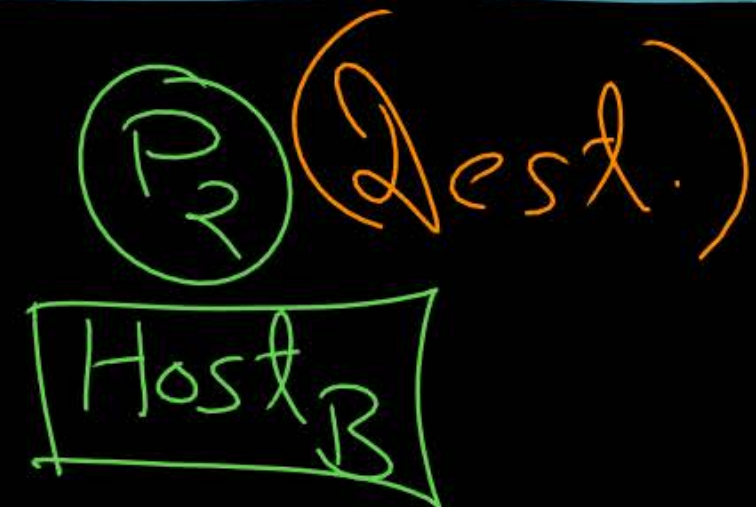
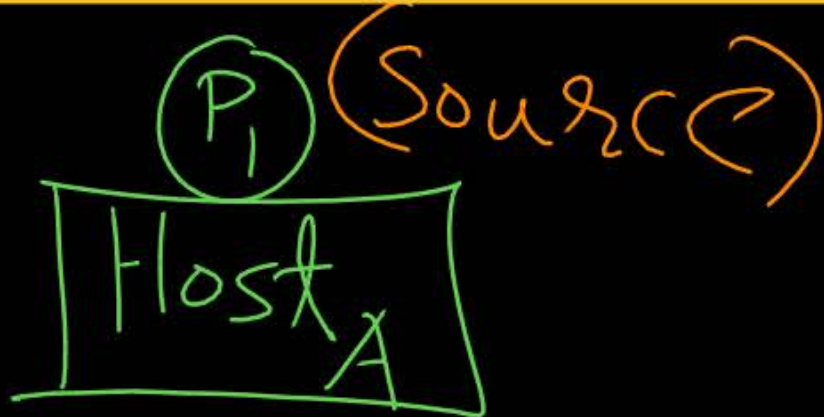
Communication Credentials

Source Port No.	(16-Bit)
Source IP Address	(32-Bit)

→ To identify P_1
Source process

Destination IP Address	(32-Bit)
Destination Port No.	(16-Bit)

→ To identify P_2
Dest. Process





2 mins Summary



Topic

Application Layer

Topic

Two Process Communication

Topic

~~Transport Layer~~



THANK - YOU

